A Cutoff-Sensitive Dispersive Model for Faraday Rotation in Transparent Media: Broadband Verdet Scaling, Line Shape, and Effective High-Frequency Reduction

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Abstract

We present a compact dispersive parameterization for Faraday rotation in transparent dielectrics based on circular birefringence within a Lorentz-type linear-response framework. The model remains fully orthodox at the level of macroscopic electrodynamics, but reorganizes the standard response in terms of a single effective high-frequency scale ω_* that governs the transparent-band approach to the ultraviolet resonance sector. Closed-form expressions are derived for the circular-index splitting $\Delta n(\omega, B)$ and the Verdet constant $\mathcal{V}(\omega)$, including damping. We show how ω_* can be interpreted as the moment-matched reduction of a multi-oscillator background, derive the transparent-limit scaling laws, and formulate a practical parameter-extraction strategy for broadband data. We further provide a synthetic recovery test and an illustrative consistency check against known wavelength trends in silica-class materials. The framework can be used as a compact parameterization to compare materials and to constrain an effective high-frequency scale implicit in the magneto-optic response.

1 Introduction

Faraday rotation is a standard magneto-optic effect in which a longitudinal magnetic field induces circular birefringence and thereby rotates the plane of polarization of transmitted light. For a monochromatic wave of angular frequency ω traversing a medium of thickness L, the rotation angle is

$$\theta(\omega) = \frac{\omega L}{2c}\,\mathrm{Re}[n_-(\omega, B) - n_+(\omega, B)]\,, \tag{1}$$

where $n_\pm$ are the refractive indices of the two circular polarization eigenmodes and c is the vacuum speed of light. We work throughout in SI units, so that the Verdet constant is

$$\mathcal{V}(\omega) = \frac{\theta(\omega)}{BL}, \tag{2}$$

with units $\mathrm{rad\,T^{-1}\,m^{-1}}$.

The standard theoretical description of Faraday rotation in transparent media is classical and well established. In the weak-field regime it is commonly derived from magnetically split Lorentz oscillators or, equivalently, from the antisymmetric part of the dielectric tensor in a magnetized medium. Historical forms include the Becquerel-type relation linking the Verdet constant to material dispersion, while textbook treatments develop the effect as circular birefringence induced by field-dependent constitutive response. The present work does *not* attempt to replace these standard descriptions.

The narrower aim of this paper is instead the following. Many transparent materials exhibit broadband Verdet data that are adequately smooth over a wide spectral band, even though microscopically the optical response may arise from several ultraviolet or interband resonances. In that regime it

is useful to ask whether one can compress the magneto-optic response into a single effective high-frequency scale that is operationally extractable from broadband data and that separates naturally from the overall response amplitude and damping width. The contribution of this paper is to show that such a reduction exists in a simple closed form, to interpret the resulting ω_* as an effective resonance centroid rather than as a new fundamental frequency, and to give a practical fitting framework for transparent-band data.

Accordingly, the novelty claim here is *not* "a new derivation of the Faraday effect." Rather, it is:

1. a minimal one-scale reduction of a potentially multi-resonance magneto-optic response;
2. an explicit moment-matching interpretation of the fitted scale ω_*;
3. closed-form line-shape expressions convenient for broadband fitting and transparent-band model discrimination.

Here we introduce a compact parameterization that isolates a single effective high-frequency scale ω_* governing the transparent-limit behavior. This scale is treated phenomenologically and may be interpreted as the moment-matched reduction of unresolved higher-frequency oscillator strength within the material response.

2 Model: circular-mode dispersion with resonance splitting

Let ω_L denote the Larmor frequency associated with the applied magnetic field. In the simplest free-electron form,

$$\omega_L = \frac{eB}{2m_e}, \tag{3}$$

where e is the elementary charge and m_e the electron mass. We model the two circular eigenmodes by

$$n_\pm^2(\omega, B) = 1 + \frac{S}{(\omega_* \pm \eta\omega_L)^2 - \omega^2 - i\gamma\omega}, \tag{4}$$

where S is an oscillator-strength parameter with dimensions of angular-frequency squared, γ is an effective damping rate, and η is a dimensionless magneto-coupling factor that absorbs effective-mass and geometry renormalizations.

Define

$$D(\omega) := \omega_*^2 - \omega^2 - i\gamma\omega. \tag{5}$$

For weak fields $\omega_L \ll \omega_*$,

$$(\omega_* \pm \eta\omega_L)^2 = \omega_*^2 \pm 2\eta\omega_*\omega_L + \mathcal{O}(\omega_L^2), \tag{6}$$

so that

$$n_\pm^2(\omega, B) \approx 1 + \frac{S}{D(\omega) \pm 2\eta\omega_*\omega_L}. \tag{7}$$

Using

$$\frac{1}{D \pm \delta} \approx D^{-1} \mp \delta D^{-2} \qquad (|\delta| \ll |D|), \tag{8}$$

we obtain

$$n_\pm^2 \approx 1 + \frac{S}{D} \mp \frac{2\eta S\omega_*\omega_L}{D^2}. \tag{9}$$

Hence

$$\Delta(n^2) := n_-^2 - n_+^2 \approx \frac{4\eta S\omega_*\omega_L}{D(\omega)^2}. \tag{10}$$

For weak circular splitting, write $n_\pm = n_0 \pm \delta n/2$, where $n_0 := n(\omega, B = 0)$. Then

$$\Delta n := n_- - n_+ \approx \frac{\Delta(n^2)}{2n_0}, \tag{11}$$

so that

$$\boxed{\Delta n(\omega, B) \approx \frac{2\eta S \omega_* \omega_L}{n_0\, D(\omega)^2}}. \tag{12}$$

3 Faraday rotation and Verdet constant

Substituting (12) into (1) gives

$$\theta(\omega) = \frac{\omega L}{2c}\, \mathrm{Re}[\Delta n(\omega, B)]\,, \tag{13}$$

hence

$$\mathcal{V}(\omega) = \frac{\omega}{2cB}\, \mathrm{Re}[\Delta n(\omega, B)]\,. \tag{14}$$

Using $\omega_L = eB/(2m_e)$, we obtain

$$\boxed{\mathcal{V}(\omega) \approx \frac{\eta e\, \omega\, S\, \omega_*}{2m_e c\, n_0}\, \mathrm{Re}\left[\frac{1}{D(\omega)^2}\right]}. \tag{15}$$

This is the central fitting expression of the paper.

4 Transparent, weak-damping limit

If $\gamma\omega \ll \left|\omega_*^2 - \omega^2\right|$, then $D(\omega) \approx \omega_*^2 - \omega^2$ is approximately real, and

$$\mathcal{V}(\omega) \approx \frac{\eta e\, \omega\, S\, \omega_*}{2m_e c\, n_0 (\omega_*^2 - \omega^2)^2}. \tag{16}$$

At zero field, (4) reduces to

$$n_0^2 - 1 = \frac{S}{\omega_*^2 - \omega^2 - i\gamma\omega}. \tag{17}$$

In the transparent weak-damping regime this becomes

$$n_0^2 - 1 \approx \frac{S}{\omega_*^2 - \omega^2}, \tag{18}$$

so we may eliminate S:

$$\boxed{\mathcal{V}(\omega) \approx \frac{\eta e\, \omega\, (n_0^2 - 1)\, \omega_*}{2m_e c\, n_0 (\omega_*^2 - \omega^2)}}. \tag{19}$$

In the deep sub-cutoff regime $\omega \ll \omega_*$,

$$\boxed{\mathcal{V}(\omega) \approx \frac{\eta e}{2m_e c}\, \frac{\omega(n_0^2 - 1)}{n_0}\, \frac{1}{\omega_*}}. \tag{20}$$

Thus the transparent-band prediction is that $\mathcal{V}(\omega)/\omega$ approaches a slowly varying quantity proportional to $1/\omega_*$, up to the known dispersion of $n_0(\omega)$.

5 What is new here: extraction of an effective high-frequency scale from broadband Verdet data

Equations (15)–(20) are intended as a practical fitting template for broadband magneto-optic measurements. The novelty claim here is not a new derivation of the Faraday effect itself, but rather a compact one-scale reduction of a potentially multi-resonance response, together with an explicit interpretation of ω_* and an operational fitting strategy.

Moment-matched interpretation of ω_*. Suppose the zero-field dielectric response is generated by several oscillator families,

$$n_0^2(\omega) - 1 = \sum_{j=1}^{N} \frac{S_j}{\Omega_j^2 - \omega^2 - i\gamma_j\omega}. \tag{21}$$

Far below the absorption sector, the weak-damping expansion is

$$n_0^2(\omega) - 1 = \sum_j \frac{S_j}{\Omega_j^2} + \omega^2 \sum_j \frac{S_j}{\Omega_j^4} + \mathcal{O}(\omega^4, \gamma_j\omega). \tag{22}$$

If we approximate this by a single effective oscillator,

$$n_0^2(\omega) - 1 \approx \frac{S_{\text{eff}}}{\omega_*^2 - \omega^2 - i\gamma\omega}, \tag{23}$$

then matching the first two even moments gives

$$\frac{S_{\text{eff}}}{\omega_*^2} = \sum_j \frac{S_j}{\Omega_j^2}, \qquad \frac{S_{\text{eff}}}{\omega_*^4} = \sum_j \frac{S_j}{\Omega_j^4}, \tag{24}$$

and therefore

$$\boxed{\omega_*^2 = \frac{\sum_j S_j/\Omega_j^2}{\sum_j S_j/\Omega_j^4}}, \qquad \boxed{S_{\text{eff}} = \frac{\left(\sum_j S_j/\Omega_j^2\right)^2}{\sum_j S_j/\Omega_j^4}}. \tag{25}$$

Thus ω_* should be read as an effective high-frequency centroid of unresolved oscillator strength, not as a claim of a new microscopic resonance.

This two-moment reduction is exact only to the extent that the transparent-band response is well represented by its first two even-power moments. If several resonance families contribute comparably over a broad spectral interval, higher moments become non-negligible and the one-scale reduction should be replaced by a multi-oscillator fit.

Fit strategy (minimal). Measure $\mathcal{V}(\omega)$ and $n_0(\omega)$, then fit either:

1. (15) with parameters $(\omega_*, \eta S, \gamma)$, or
2. (19) in the transparent limit with reduced parameters (ω_*, η).

At leading order, broadband transparent-band data constrain the product ηS more robustly than η and S separately.

Order of magnitude of η. The dimensionless factor η absorbs effective-mass and geometry renormalizations. In ordinary dielectric applications one expects it to be at most order unity unless unusually strong renormalization is intended. In practice, the illustrative silica-class estimate in Sec. 8 gives $\eta \sim 10^{-2}$–10^{-1} for $\omega_* \sim 10^{16}\,\text{s}^{-1}$, so the model does not require anomalously large magneto-coupling.

Falsifiable signatures.

1. **Band dependence of extracted ω_*:** if ω_* is a genuine effective reduction of unresolved high-frequency structure, then fits over moderately shifted transparent windows should return approximately stable values of ω_*, whereas strongly separated windows need not.

2. **Low-frequency scaling:** in a sufficiently transparent band, $\mathcal{V}(\omega)/\omega$ should approach a slowly varying quantity proportional to $1/\omega_*$.

3. **Damping-controlled line shape:** the factor $\mathrm{Re}[D^{-2}]$ predicts a characteristic sign-preserving enhancement near weak resonances; systematic deviations constrain whether the one-scale template is adequate.

4. **Need for multi-oscillator extension:** structured residuals across a broad band indicate that (21), rather than the one-scale reduction, is required.

6 Synthetic recovery test

To verify that the reduced template is operationally identifiable, we generated synthetic Verdet data from Eq. (15) on a uniform wavelength grid spanning 193–1550 nm, using the input parameters

$$\omega_*^{\mathrm{in}} = 1.57 \times 10^{16}\ \mathrm{s}^{-1}, \qquad (\eta S)^{\mathrm{in}} = 2.30 \times 10^{32}\ \mathrm{s}^{-2}, \qquad \gamma^{\mathrm{in}} = 5.0 \times 10^{13}\ \mathrm{s}^{-1}. \tag{26}$$

Gaussian relative noise was then added, and the full three-parameter model was refit by nonlinear least squares.

Representative synthetic values from the input model are

$$\mathcal{V}(1550\,\mathrm{nm}) \approx 15.4\ \mathrm{rad\,T^{-1}\,m^{-1}}, \tag{27}$$

$$\mathcal{V}(632.8\,\mathrm{nm}) \approx 39.0\ \mathrm{rad\,T^{-1}\,m^{-1}}, \tag{28}$$

$$\mathcal{V}(405\,\mathrm{nm}) \approx 65.5\ \mathrm{rad\,T^{-1}\,m^{-1}}, \tag{29}$$

$$\mathcal{V}(193\,\mathrm{nm}) \approx 267\ \mathrm{rad\,T^{-1}\,m^{-1}}. \tag{30}$$

A Monte Carlo recovery test gives the following mean recovered parameters:

Noise level	ω_*^{fit}	$(\eta S)^{\mathrm{fit}}$	γ^{fit}
1%	$1.5688(53) \times 10^{16}$	$2.293(33) \times 10^{32}$	$2.32(247) \times 10^{14}$
3%	$1.566(20) \times 10^{16}$	$2.282(120) \times 10^{32}$	$2.19(245) \times 10^{14}$

Here the parenthetical uncertainties denote one standard deviation in the last quoted digits across the Monte Carlo ensemble.

The main practical conclusion is that ω_* and the amplitude combination ηS are recovered robustly, while γ is much more weakly constrained unless the fitted band extends sufficiently close to the absorption shoulder. This behavior is consistent with the regime discussion above and clarifies which parameters are realistically extractable from transparent-band data.

7 Regime validity and limitations

The model is a compact linear-response parameterization suitable for weak fields and transparent propagation bands. It is not intended for strongly absorbing regions, dense plasmas, or regimes requiring several comparable oscillator families unless explicitly extended.

More specifically:

1. the weak-field expansion requires $\eta\omega_L \ll \omega_*$ and $|2\eta\omega_*\omega_L| \ll |D(\omega)|$;

2. the transparent-limit formulas assume $\gamma\omega \ll \left|\omega_*^2 - \omega^2\right|$;

3. the one-scale reduction is appropriate only when the unresolved ultraviolet/interband background can be compressed meaningfully into the first two even moments of the response;

4. fitted values of ω_* should be interpreted operationally, not as unique microscopic transition frequencies.

8 Illustrative silica-class consistency check

A full material fit is beyond the scope of the present paper, but a simple order-of-magnitude check is useful. Reported silica-class values include approximately

$$\mathcal{V}(1523\,\mathrm{nm}) \approx 0.54\ \mathrm{rad\,T^{-1}\,m^{-1}}, \qquad \mathcal{V}(632.8\,\mathrm{nm}) \approx 3.61\ \mathrm{rad\,T^{-1}\,m^{-1}},$$

for silica-fiber-class measurements, and

$$\mathcal{V}(193\,\mathrm{nm}) = 70.1\ \mathrm{rad\,T^{-1}\,m^{-1}}$$

for synthetic quartz / silica glass in the deep ultraviolet.

Using the transparent-limit form

$$\mathcal{V}(\omega) \propto \frac{\omega}{\omega_*^2 - \omega^2}, \tag{31}$$

the ratio

$$\frac{\mathcal{V}(193\,\mathrm{nm})}{\mathcal{V}(1523\,\mathrm{nm})} \approx \frac{70.1}{0.54} \approx 1.30 \times 10^2$$

is reproduced at the order-of-magnitude level by an effective scale

$$\omega_* \sim 1.0 \times 10^{16}\ \mathrm{s^{-1}} \qquad (\lambda_* \sim 190\ \mathrm{nm}).$$

This places the extracted scale in the expected ultraviolet sector and shows that the one-scale reduction is at least numerically plausible for silica-class trend analysis.

This comparison is intentionally only illustrative: it mixes data from silica-class bulk and fiber implementations, so it should not be read as a precision fit. Its purpose is only to show that the magnitudes and wavelength trend implied by the model are realistic.

9 Conclusion

We derived a minimal dispersive model for Faraday rotation that isolates an effective high-frequency scale ω_* in the circular-birefringent response. The result yields closed-form expressions for Δn and $\mathcal{V}(\omega)$ with damping, provides an explicit moment-matching interpretation of ω_*, and gives a practical route to constrain this scale from broadband Verdet data.

The present contribution is not a new microscopic theory of the Faraday effect. Rather, it is a compact reduced parameterization of standard magneto-optic dispersion. The synthetic recovery test shows that ω_* and the amplitude combination ηS are robustly identifiable in transparent-band fits, while the silica-class order-of-magnitude check indicates that the extracted scale lies in a physically sensible ultraviolet range. The framework therefore offers a simple, bounded, and operationally useful way to compare materials and quantify the high-frequency structure implicit in magneto-optic response.

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